

Fundamentals of Inference

SURV 740 (UMD)

SURVMETH 740 (UMICH)

Fall 2025

Fridays, 1-4 PM

(1070 ISR / 1208 Lefrak)

First Class: Friday, August 29, 2025

Last Class: Friday, December 5, 2025

Professor: **Dr. Yan Li**
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Joint Program in Survey Methodology and
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Invited Guest Lecturer: Dr. Barry Graubard
Scientist Emeritus
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Office Hours: by appointment

Course Description: This course is one of the three fundamental courses required by all students in the master's programs at JPSM and MPSDS and focuses on the fundamentals of statistical inference in the finite population setting.

Course Materials: Reading material as provided (see schedule).

Pre-requisites: SurvMeth 685/686, Surv 615/616, or Biostat 601/602.

Course Goals:

This course is designed to give an overview and review fundamental ideas of making inferences about finite populations. It will emphasize basic principles of probability sampling; focus on differences between making predictions and making inferences; explore the differences between randomized study designs and observational studies; consider model-based vs. design-based analytic approaches; review techniques designed to improve efficiency using auxiliary information; introduce causal inference; and consider non-probability sampling and related inferential techniques.

Course Requirements:

Weekly reading assignments will be provided. Class will be a mixture of lecture and discussion; students should read the assignments and come prepared to discuss the material during class.

Homework assignments will be provided approximately every two to three weeks during the semester. **Homework is to be submitted electronically via the course website (Canvas) by 11:59 PM on the due date.** Solutions may be handwritten or typed, but must be submitted in a single PDF format file, with your name and homework number at the top of the first page, and page numbers at the bottom of each page. Files must be submitted with a standard naming convention: ‘Surname First-Initial HW #.pdf’ (for example, ‘Li Y HW 1.pdf’).

Without prior arrangements, homework submitted after the due date will be considered late, and 10% of the points will be deducted for each day late. Study groups are useful, and encouraged, for preparing answers to homework exercises; however, group answers are not acceptable. Each student must submit their own individual homework solutions.

There will be two open-book, open-notes in-class examinations. Each will primarily cover all material discussed since the previous exam. Students are advised to bring a calculator to the examinations; use of smartphones and laptops will not be allowed.

Grading will be as follows:

40% Homework
30% Exam I
<u>30% Exam II</u>
100%

Diversity, Equity, and Inclusion:

The University of Michigan and the University of Maryland seek to create and disseminate knowledge. Utilizing the institutional civility codes, we will promote the values of diversity, equity, and inclusion, both inside and outside our classrooms. To this end, the University of Michigan and the University of Maryland uphold the expectations that all course participants will acknowledge diverse experiences in the classroom, create

environments that encourage equitable classroom participation, and ensure that students and faculty abide by their institution's policies and procedures.

Academic Integrity:

The faculty believe that the conduct of a student registered or taking courses in their respective Universities should be consistent with that of a professional person. Courtesy, honesty, and respect should be shown by students toward faculty members, guest lecturers, administrative support staff, community partners, and fellow students. Similarly, students should expect faculty to treat them fairly, showing respect for their ideas and opinions and striving to help them achieve maximum benefits from their experience.

Student academic misconduct refers to behavior that may include plagiarism, cheating, fabrication, falsification of records or official documents, intentional misuse of equipment or material (including library materials), and aiding and abetting the perpetration of such acts. Please visit <https://rackham.umich.edu/academic-policies/section8/> for the full policy on roles and responsibilities for the University of Michigan graduate students, and <https://policies.umd.edu/academic-affairs/university-of-maryland-code-of-academic-integrity> for the University of Maryland Code of Academic Integrity.

Student Accommodations:

Students should speak with their instructors before or during the first week of classes regarding any special needs. Students seeking academic accommodations at the University of Michigan should register with Services for Students with Disabilities (SSD). SSD arranges reasonable and appropriate academic accommodations for students with disabilities. Please visit <http://ssd.umich.edu/> for more information on student accommodations. University of Maryland students should contact the Access and Disability Service: <http://www.counseling.umd.edu/ads/>.

Students who expect to miss classes, examinations, or other assignments as a consequence of their religious observance shall be provided with a reasonable alternative opportunity to complete such academic responsibilities. It is the obligation of students to provide faculty with reasonable notice of the dates of religious holidays on which they will be absent. Please visit <https://provost.umich.edu/resources-policies/calendars/> or <https://provost.umd.edu/calendar> for the complete University policies at the respective institutions.

Course Reading List:

Readings will be referenced by number each week in the syllabus. The readings are available at the course website.

Required Readings (Causal Inference):

1. Baker R., Brick J.M., Bates N.A., Battaglia M., Couper M.P., Dever J.A., Gile K.J., Tourangeau, R. (2013). Summary report of the AAPOR task force on non-probability sampling. *Journal of Survey Statistics and Methodology*, 1:90-143.
2. Cornesse, C., Blom, A. G., Dutwin, D., Krosnick, J. A., De Leeuw, E. D., Legleye, S., & Struminskaya, B. (2020). A review of conceptual approaches and empirical evidence on probability and nonprobability sample survey research. *Journal of Survey Statistics and Methodology*, 8(1), 4-36.
3. D'Agostino Jr., R.B. (1998). Tutorial in biostatistics: Propensity score methods for bias reduction in the comparison of a treatment to a non-randomized control group. *Statistics in Medicine*, 17, 2265-81.
4. Elliott M.R. and Valliant R. (2017). Inference for nonprobability samples. *Statistical Science*, 32:249-264.
5. Hays R.D., Liu H., Kapteyn A. (2015). Use of internet panels to conduct surveys. *Behavior Research Methods*, 47:685-690.
6. Hernán M.A., Robins J.M. (2017). *Causal Inference*. Boca Raton: Chapman & Hall/CRC, forthcoming. Preprint available at <http://www.hsph.harvard.edu/miguel-hernan/causal-inference-book/>. Last accessed Jan. 29, 2018 (Chapters 1-2)
7. Jackson, M., Cox D.R. (2013). The principles of experimental design and their application in sociology, *Annual Review of Sociology*, 39:27-49.
8. Johnson T.P., Smith T.W. (2017). Big Data and Survey Research: Supplement or Substitute?. In *Seeing Cities Through Big Data* (pp. 113-125). Springer
9. Joffe, M.M. Ten Have, T.R., Feldman, H.I., Kimmel, S.E. (2004). model selection, confounder control, and marginal structural models, *The American Statistician*, 58:272-279.
10. Lee, S. (2006). Propensity Score Adjustment as a Weighting Scheme for Volunteer Panel Web Surveys. *Journal of Official Statistics*, 22(2), 329-349.
11. Lohr S.L., Raghunathan T.E. (2017). Combining survey data with other data sources. *Statistical Science*, 32:293-312.
12. Rubin, D.B. (2006). Causal inference through potential outcomes and principal stratification: Application to studies with “censoring” due to death, *Statistical Science*, 21: 299-309.
13. Rubin, D.B. (2007). The Design versus the analysis of observation studies for causal effects: Parallels with the design of randomized trials. *Statistics in Medicine*, 26, 20-36.
14. Stuart, E.A. (2010). Matching Methods for Causal Inference: A Review and a Look Forward. *Statistical Science: A Review Journal of the Institute of Mathematical Statistics*, 25(1), 1.

Required Readings (Randomization-based and Model-based Inference):

15. Brewer, K.R.W. (1963), Ratio estimation and finite populations: some results deducible from the assumption of an underlying stochastic process, *Australian Journal of Statistics*, 5, 93-105, <https://doi.org/10.1111/j.1467-842X.1963.tb00288.x>
16. Brewer, K. and T. G. Gregoire (2009). Chapter 1 - Introduction to Survey Sampling.

- Handbook of Statistics*. C. R. Rao, Elsevier. 29: 9-37.
17. Hansen, M. H., Madow, W. G. and Tepping, B. J. (1983). An Evaluation of Model-Dependent and Probability-Sampling Inferences in Sample Surveys, *J. American Statist. Assoc.* 78 776 - 793 (with discussion).
 18. Holt, D. and Smith, T. M. F. (1979). Poststratification, *J. Royal Statist. Soc. A* 142 33 - 46.
 19. Kalton, G. (2019). Developments in survey research over the past 60 years: A personal perspective. *Int. Stat. Rev.*, 87, S10-S30.
 20. Little, R. J. A. (1993). Post-Stratification: a Modeler's Perspective, *J. American Statist. Assoc.* 88 1001 - 1012.
 21. Little, R. J. A. (2004). To Model or Not to Model? Competing Modes of Inference for Finite Population Sampling, *J. American Statist. Assoc.* 99 546 - 556.
 22. Little, R. (2008), Weighting and prediction in sample surveys, *Calcutta Statistical Association Bulletin*, Vol. 60, 1-17. It is not necessary to read the discussion.
 23. Little, R. J. (2006). "Calibrated Bayes: a Bayes/Frequentist Roadmap." American Statistician **60**(3): 213.
 24. Rao, J. N. K. (1999). Some current trends in sample survey theory and methods (with discussion). *Sankhyā* Ser. B, 61, 1-57.
 25. Royall, R.M. (1970). On finite population sampling theory under certain linear regression models. *Biometrika*, 57, 377-387.
 26. Sarndal, C.-E. (2007). The calibration approach in survey theory and practice. *Survey Methodology*, 33, 99-111
 27. Korn, EL., & Graubard, BI. (1995). Analysis of Large Health Surveys: Accounting for the Sampling Design. *Journal of the Royal Statistical Society. Series A (Statistics in Society)*, 158(2), 263-295

Course Schedule:

Week	Date	Content	Instructor
1	08/29/2025	Design and Analysis of Randomized Experiments for Causal Inferences Hernan and Robins (2017) Ch 1-2; Jackson and Cox (2013) HW1 assigned	YL
2	09/05/2025	Design and Analysis of Observational Studies for Causal inferences Hernan and Robins (2017) Ch 3; Rubin (2007)	YL
3	09/12/2025	Propensity Scores D'Agostino (1998); Stuart (2010) HW1 due (11:59 PM); HW2 assigned	YL
4	09/19/2025	Other Analysis Tools for Causal Inference: Marginal Structural Models Joffe et al. (2004); Rubins et al. (2000)	YL
5	09/26/2025	Other Analysis Tools for Causal Inference: Principal Stratification Rubin (2006) HW2 due (11:59PM); HW3 assigned	YL
6	10/03/2025	Non-probability sample analysis Baker et al. (2013); Cornesse et al. (2020); Johnson and Smith (2017); Elliott & Valliant (2017); Lohr & Raghunathan (2017); Lee (2006) HW3 due (11:59 PM)	YL
7	10/10/2025	Mid-Term Exam (Exam I)	
8	10/17/2025	Randomization Inference RR: #16, #19, #24 HW4 assigned	BG YL
9	10/24/2025	Randomization Inference RR: #15, #17	BG YL
10	10/31/2025	Model-based Inference RR: #21, #23, #25, #27 HW4 due (11:59 PM); HW5 assigned	BG YL
11	11/07/2025	Model-based Inference	BG YL
12	11/14/2025	Domains and Poststratification RR: #18, #20, #22 HW5 due (11:59 PM)	BG YL
13	11/21/2025	Model-assisted Inference RR: #26	BG YL
14	11/28/2025	Thanksgiving Holiday: No Class	
15	12/05/2025	Final Exam (Exam II)	